$$\alpha = \sqrt{\frac{4 \times 10^{-4}}{10^{-2}}}$$

$$= \sqrt{4 \times 10^{-2}}$$

$$= 2 \times 10^{-1}$$

$$= 0.2$$

For 1×10^{-4}M acid,

$$\alpha = \sqrt{\frac{4 \times 10^{-4}}{10^{-4}}}$$

$$= 2$$

i.e, When the dilution increases by 100 times, (Concentration decreases from 1×10^{-2}M to 1×10^{-4}M), the dissociation increases by 10 times.

Thus, we can conclude that, when dilution increases, the degree of dissociation of weak electrolyte also increases. This statement is known as Ostwald's dilution Law.

The concentration of $H^+(H_3O^+)$ can be calculated using the K_a value as below.

$$[H^+] = \alpha C \qquad \text{(Refer table)} \quad .....(8.16)$$

Equilibrium molar concentration of $[H^+]$ is equal to αC

$$\therefore [H^+] = \left(\sqrt{\frac{K_a}{C}}\right) C \qquad [\because \text{ equation (8.15)}]$$

$$= \sqrt{\frac{K_a C^2}{C}}$$

$$[H^+] = \sqrt{K_a C} \qquad .....(8.17)$$

Similarly, for a weak base

$$K_b = \alpha^2 C \quad \text{and} \quad \alpha = \sqrt{\frac{K_b}{C}}$$

$$[OH^-] = \alpha C$$

(or)

$$[OH^-] = \sqrt{K_b C} \qquad .....(8.18)$$

Example 8.4

A solution of 0.10M of a weak electrolyte is found to be dissociated to the extent of 1.20% at 25°C . Find the dissociation constant of the acid.

Given that $\alpha = 1.20\% = \frac{1.20}{100} = 1.2 \times 10^{-2}$ $K_a = \alpha^2 c$

$= (1.2 \times 10^{-2})^2 \ (0.1) = 1.44 \times 10^{-4} \times 10^{-1}$

$= 1.44 \times 10^{-5}$

XII U8-Ionic equilibrium.indd 14 16-12-2022 19:33:53